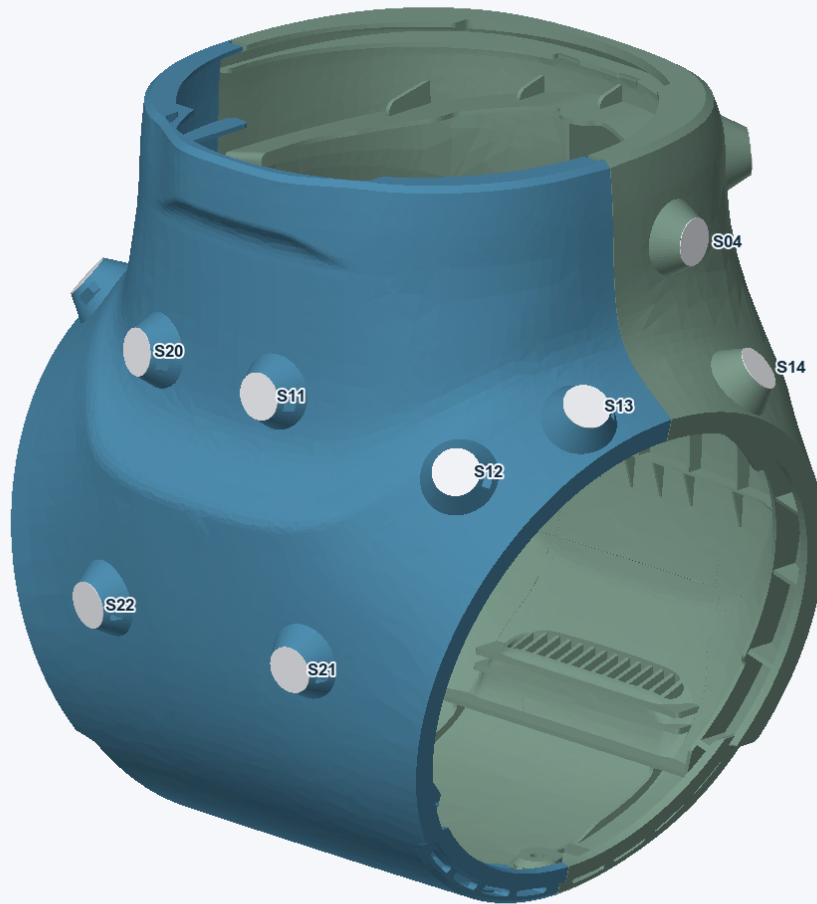


A3 dual-mode marker shell v3

12 retained asymmetric flat stations / 12 mm stickers or balls / geometry checked

A3 v3 / 12 retained flat-face dual-mode stations

Old mounts removed | printable mesh geometry checked | physical validation pending



Blue/green: one printable part per half. White: 12 mm reflective stickers (0.20 mm thickness).

Flat-face revision: 12 mm round stickers lie on flat 12 mm station tops. No raised rim or recessed sticker pocket remains. A small flush pin supports the sticker over the central screw bore; film plus adhesive is 0.20 mm.

Ball mode: remove sticker and central pin to expose the same-axis screw bore. M3 captive-nut hardware is provisional. Neither the exposed sticker nor a protruding ball has a claimed fall-survival rating.

Old features removed: all ten obsolete external bosses and screw bores are replaced by smooth local skin. Structural fasteners and all 12 retained station screw bores remain. The other twelve v3 stations and their holes are also removed. Upper internal reliefs and ribs are retained; old cavities are closed locally at the restored wall.

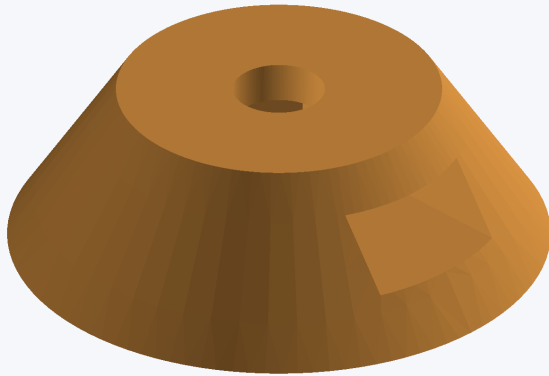
Printable geometry: both fused halves pass closed-manifold, winding and two independent intersection checks. Marker positions and axes are unchanged and checked on the exported meshes. Physical fit, printing process, ball hardware and crash survival remain unverified.

One station, two marker modes

Provisional M3 hardware / no hole drilled through the base shell interior

Flat 12 mm top / central screw bore preserved

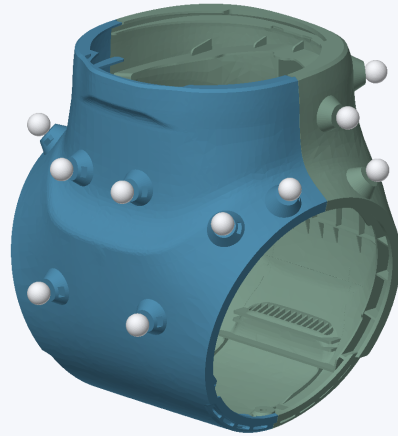
No raised rim or sticker pocket | 3.4 mm bore | M3 hardware provisional | pin removed



Local hardware-fit illustration. Ball thread, nut and plug fit must be validated before shell release.

Ball mode / illustrative 12 mm spheres

Old mounts removed | printable mesh geometry checked | physical validation pending



Ball socket thread and actual ball-centre offset require hardware verification. Ball mode is not crash-protected.

Station: a flat 12.0 mm outside-diameter top with no raised guard or sticker recess. Its tapered 20.0 mm supporting foot stays below the top plane. The central screw bore is 3.4 mm diameter and 4.6 mm blind depth below the seat.

Nut pocket: nominal M3 hex nut, 5.5 mm across flats and approximately 2.4 mm thick. Pocket clearance is 5.8 mm across flats by 2.65 mm high, with the pocket top 1.5 mm below the seat. It loads from the outside through a side slot. Confirm the actual nut and retention method.

Sticker installation, after fit checks: seat the 3.45 mm diameter, 1.30 mm long central pin fully flush, then centre one 12 mm sticker across the entire top. The nominal interference is a trial fit. Verify retention and a level face; do not force or sand the optical support plane.

Ball installation, after fit checks: remove pin and sticker, install the confirmed ball stud and retain the nut. Candidate stud penetration is limited to 4.0 mm below the seat. Verify actual thread engagement and seating; do not drill deeper or force a stud into the blind floor.

Different centres: the ball sphere centre is not the sticker surface centre. The displayed ball spheres use an illustrative 6 mm offset only. Measure the installed ball, stud and spacer arrangement before creating the ball-mode table and Motive rigid body. Do not mix the two calibration tables.

Marker translations to pelvis_link

Nominal CAD coordinates / metres / mount seat is NOT ball sphere centre

The sticker centre includes 0.20 mm film and adhesive above the flat station top. Seat coordinates locate the concentric screw axis at that plane. The retained stations keep their original positions and orientations; the removed stations are omitted. Values are rounded to 0.000001 m, not an accuracy claim.

ID	Sticker X	Sticker Y	Sticker Z	Seat X	Seat Y	Seat Z
S04	-0.023681	0.072053	-0.061631	-0.023648	0.071856	-0.061636
S05	-0.067166	0.047266	-0.054582	-0.067006	0.047146	-0.054589
S11	0.068681	0.013753	-0.093372	0.068482	0.013736	-0.093385
S12	0.055718	0.068988	-0.102706	0.055579	0.068885	-0.102807
S13	0.024198	0.082666	-0.090054	0.024146	0.082537	-0.090198
S14	-0.035526	0.082456	-0.094529	-0.035453	0.082323	-0.094659
S16	-0.086616	-0.018974	-0.113904	-0.086419	-0.018963	-0.113934
S18	-0.022710	-0.083580	-0.089898	-0.022665	-0.083449	-0.090042
S19	0.034664	-0.079647	-0.092672	0.034583	-0.079518	-0.092801
S20	0.066488	-0.030622	-0.093038	0.066293	-0.030580	-0.093060
S21	0.079132	0.034881	-0.156415	0.078934	0.034855	-0.156395
S22	0.079113	-0.034871	-0.156403	0.078915	-0.034846	-0.156383

Machine-readable documents: the supplied XLSX and two CSV tables include all 12 retained translations, qx/qy/qz/qw quaternions and outward normals. The transform maps each marker-local frame into pelvis_link. Local +z points outward; round-marker roll is a CAD convention.

Ball optical centres: not yet known. Add the measured installed ball-centre offset along the outward normal to the seat position. Do not treat the seat as the sphere centre or assume a 6 mm offset from diameter alone.

CAD registration and verification

Calculated from source datums / not a live mocap or physical-fit calibration

Frame source: the original STEP has no named robot root. Ten existing nominal marker coordinates define pelvis_link; an existing calibration record identifies the same frame. The old live P1-to-pelvis transform is not reused.

Direct CAD evidence: ten original socket-end annuli were read from the latest source STEP. Each has concentric 1.6 mm and 4.0 mm radii and area 42.223005 mm². Front and rear outward normals were verified. Named correspondences retain the reference front/back and left/right identities.

Joint solve: $q_{\text{reference}} = R(p_{\text{socket}} + h n) + t$, in millimetres. A proper rotation, translation and common original marker stand-off were fitted simultaneously (seven parameters, 30 coordinates). Stand-off was not imposed; it converged to 6 mm. This recovered value is only for the original registration datums.

CAD-to-pelvis mapping:

$$x_m = \text{CAD_Z} / 1000$$

$$y_m = \text{CAD_X} / 1000$$

$$z_m = (\text{CAD_Y} - 145.800095037096) / 1000$$

The cyclic axis permutation is right-handed, with determinant +1.

Checks passed: maximum fitted datum residual 4.122e-13 mm; maximum leave-one-out residual 4.673e-13 mm; Jacobian rank 7/7. The archived 24-station additions STEP was checked before pruning. All 12 retained flat tops and open bores were checked again on the regenerated STL meshes. The archived analytic STEP seat discrepancy was 9.995e-12 mm; current STL top-plane and bore checks pass within 0.001 mm. These are different checks, not a physical accuracy claim.

Table checks: workbook values agree with the independent geometry calculation. A +0.10 mm film-thickness change shifts sticker centres by 0.10 mm along each normal and leaves mount seats unchanged; the published input was restored to 0.20 mm.

Meaning of verification: the tiny residuals demonstrate shared nominal CAD geometry and numerical agreement, not sub-micrometre physical accuracy. See CAD_registration.md for all ten datums, exact source hashes and reproducibility details. Manufacturing, mounting offsets and physical robot placement remain unverified.

Printing and validation checklist

Mesh geometry passes / printed fit and operational qualification still required

1 / Use the supplied mesh masters. Print half_a and half_b separately from printable/. Each is one fused, closed, consistently oriented solid. Independent triangle checks and FreeCAD detect zero self-intersections. The earlier invalid STEP and partial-repair previews are not manufacturing masters.

2 / Keep scale and frames explicit. STL units are millimetres: import at 100%, never auto-scale to the bed. The supplied OpenSCAD file shows an upright assembly in F5; choose half_a or half_b for F6/export. Bed orientation changes only the print pose, not the CAD-frame marker table.

3 / Inspect the sliced layers. Half footprints are approximately 176 x 194 mm before brim/support margins. The open side can face the bed using the OpenSCAD placement. Choose printer/material-specific supports and check every thin rib, bridge, nut pocket and screw bore. Keep support scars away from the flat optical tops.

4 / Watch the local thickness limit. Most restored sites sample near 2.5 mm wall thickness; the f1 patch has an approximately 1.50 mm local edge gap in a vertical-ray audit. Ensure the slicer retains it. This is not a certified global minimum wall thickness or impact rating. Check the printed shell before robot use.

5 / Check hardware and the robot. Confirm ball thread, stud reach, nut retention and flush-pin fit. Do not force an interference pin. Check structural screws, cable access, ventilation, seams and full joint travel on a secured robot. Physical fit and crash survival have not been tested.

6 / Validate optical tracking and calibration. Test full 360-degree coverage, front/rear blockage and grazing angles with the real cameras and limbs. Retained table coordinates are unchanged; old 24-station receipts are rejected. Generate a fresh live receipt for the installed 12-station layout. Measure actual ball-centre offsets before using ball mode.

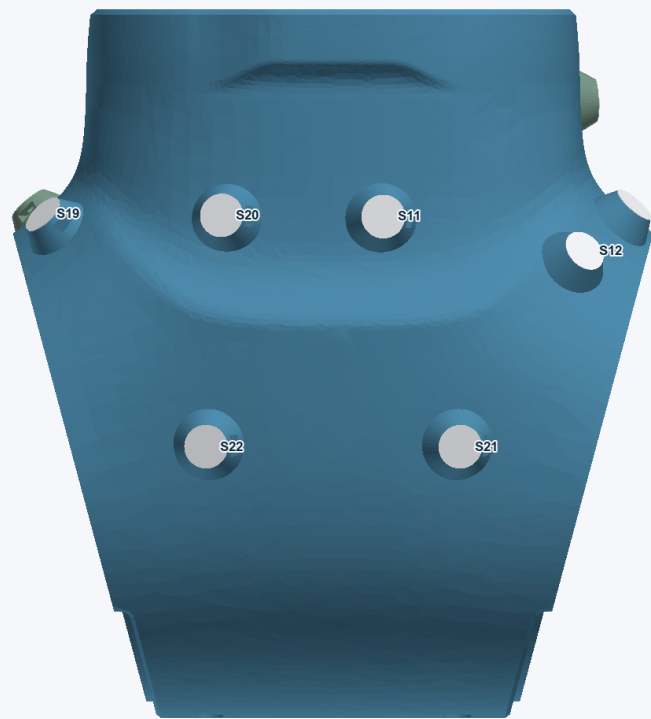
Evidence: all 588 probes at the twelve removed-station footprints agree with the smooth base shells within 0.001 mm. All 12 retained bores remain open. At 57,370 CAD reference points, maximum distance increases versus the prior v3 release are 0.00000103 mm (front) and 0.00000171 mm (rear). These are sampled checks, not physical fit certification. Full scope and hashes are in printability_validation.json.

Station identification

Original Sxx IDs preserved / 12 retained stations / retain the chosen marker-mode calibration

Shell CAD +Z half / sticker mode

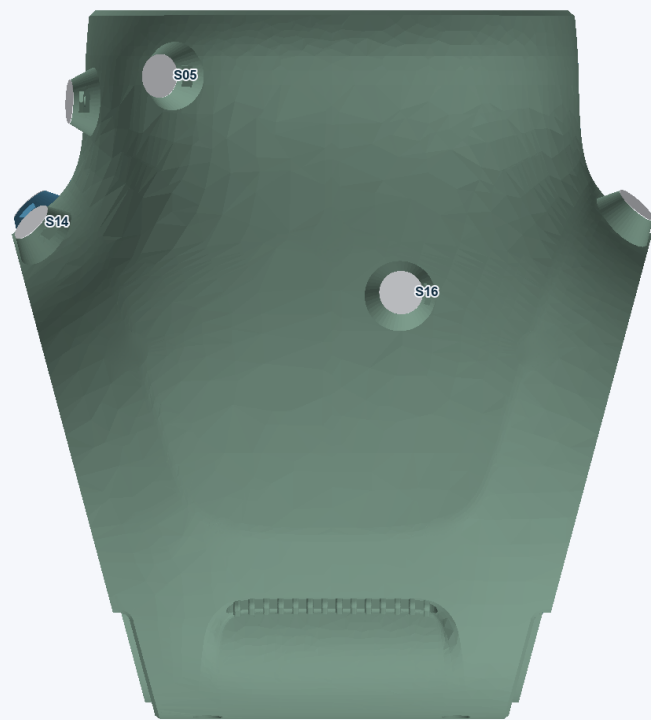
Old mounts removed | printable mesh geometry checked | physical validation pending



Blue/green: one printable part per half. White: 12 mm reflective stickers (0.20 mm thickness).

Shell CAD -Z half / sticker mode

Old mounts removed | printable mesh geometry checked | physical validation pending



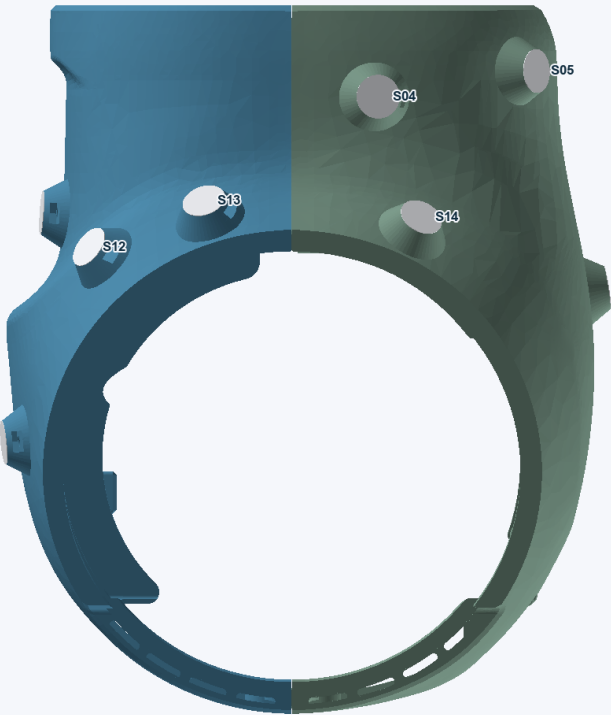
Blue/green: one printable part per half. White: 12 mm reflective stickers (0.20 mm thickness).

Side station identification

Robot left and right views / completes the retained twelve-station map

Robot left / CAD +X / sticker mode

Old mounts removed | printable mesh geometry checked | physical validation pending



Blue/green: one printable part per half. White: 12 mm reflective stickers (0.20 mm thickness).

Robot right / CAD -X / sticker mode

Old mounts removed | printable mesh geometry checked | physical validation pending



Blue/green: one printable part per half. White: 12 mm reflective stickers (0.20 mm thickness).

Pelvis-aligned OptiTrack asset

P1 local frame targets pelvis_link / expected default correction is identity

Import: motive_asset/A3_v3_12_stickers.motive now contains 12 retained markers in a rigid body named **P1**, streaming ID 9. Its origin is aligned to the nominal CAD pelvis_link datum. The matching CSV is a coordinate companion; use the native .motive file for import. The unchanged source/A3.motive retains the original 16-point arbitrary frame.

Intended ROS axes: X forward, Y left, Z up. Expected P1-to-pelvis_link correction:

Translation (m): [0, 0, 0]

Quaternion (x, y, z, w): [0, 0, 0, 1]

motive_asset/P1_to_pelvis_link_default.json records this offline expectation, not an approved runtime receipt.

Required axis settings: set Motive Streaming / Up Axis to **Z Up**. Keep the bridge setting topics.rigid_body_markers.modeldef_y_up_to_z_up = true. Native asset offsets are stored as [pelvis_x, pelvis_z, -pelvis_y], in Motive Y-up coordinates. The existing MODELDEF conversion (x,y,z) to (x,-z,y) yields the ROS frame. No extra pose rotation or old pivot correction is needed by default.

Preserved geometry: all measured markers undergo one proper rigid transform. Pair distances change by less than 8e-13 m after serialization; the points are not replaced by ideal CAD coordinates. Diameters, constraint IDs, marker order and graph/stick pair values remain preserved. CAD tables and printable meshes are unchanged.

Offline verification: the production CAD solver recovers identity from the exported, axis-converted asset to numerical precision: translation below 1e-10 m, quaternion-vector components below 1e-9. Marker-to-CAD mismatch remains 1.395 mm RMS / 1.853 mm maximum. These figures do not certify physical pelvis placement or Motive import behavior.

Safe import: test in an isolated project first because the source body UUID is retained and may replace an existing body. Verify P1, 12 markers and unique ID 9; save and re-export. Do not use Reset Location, Reset Orientation or recreate the body from visible markers: these can undo the custom pivot/axes. The asset cannot configure the Motive streaming Up Axis for you.

Before operation: restart NatNet after import and run fresh live calibration as verification. Expect an approximately identity local correction. Investigate a substantial offset; do not force a receipt to identity or set approved=true manually. Pre-alignment 12-point receipts are rejected by the new asset-frame revision. World/table alignment is separate, and the live world-to-P1 pose must still move with the robot.

Asset correspondence and evidence

Original station IDs preserved / captured-point evidence, not all-pose visibility

Retained: S04, S05, S11, S12, S13, S14, S16, S18, S19, S20, S21, S22. Removed original Points 3, 9, 13 and unmatched Point 15. All retained stations match both supplied source files; a static asset does not establish continuous visibility.

Reframe calculation: $a = R p + t$ fits nominal pelvis points p to the original measured asset points a . The new measured pelvis coordinates are $\text{transpose}(R) (a - t)$; native storage then applies $[x,z,-y]$. No scale, reflection or centroid-origin reset is used.

Traceability: documents/optitrack_asset_correspondence.json records the full matrices, hashes, source observations, aligned native coordinates and ROS coordinates. Station IDs are not renumbered; native member IDs are 1-12 in retained source order, not CAD-table order.

Station	Source Point	Member ID	CAD residual (mm)
S04	10	8	0.858511
S05	14	11	1.107767
S11	4	3	1.330930
S12	5	4	1.224260
S13	16	12	1.646311
S14	12	10	0.765856
S16	8	7	1.191584
S18	11	9	1.556395
S19	1	1	1.595486
S20	2	2	1.802336
S21	7	6	1.853359
S22	6	5	1.324662

Import/live status: native Motive import/re-export and physical live calibration have not been performed. Numerical alignment is against the nominal CAD pelvis datum, not an independently measured robot datum.

Official references:

OptiTrack: Motive Data Streaming - Up Axis

OptiTrack: Rigid Body Tracking - Pivot and Orientation controls

The exact MODELDEF (x,-z,y) mapping follows the repository field-verified convention and still requires confirmation with the current Motive 3.5 stream.